

Coding & Solution

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/shaikmubi224/OptiCrop.git
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	Crop Recommendation, Random Forest Prediction, Data Analysis, Preprocessing, Responsive UI, Crop Information Display
Pending / Incomplete Features	Weather API Integration, Fertilizer Recommendation, Disease Prediction
Setup / Run Instructions	1. Create virtual environment. 2. Install requirements using <code>pip install -r requirements.txt</code> . 3. Run <code>python app.py</code> . 4. Open http://127.0.0.1:5000/ in a browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project follows a modular architecture with separate folders for analysis, preprocessing, machine learning models, static resources, templates, and saved models. The application has been tested successfully, integrated with GitHub version control, and uses a Random Forest model with 99.32% prediction accuracy for crop recommendation.